

Sebastian Monzón

Software Engineer & Independent Researcher

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RESEARCH INTERESTS

Multimodal Interaction, Computational Perception, Computational Sensing & Design

EDUCATION

B.S. in Computer Science (Honors)

University of Massachusetts Amherst, [CGPA 3.68/4.0](#)

Amherst, MA

09/2019 - 05/2023

- Commonwealth Honors College Thesis
- Minor in Business

AWARDS

NSF Fellow (CSGrad4US)

\$159,000 covering 3 years over a 5-year fellowship period starting Fall 2026.

Scholarships

Melvin Howard Scholarship
Ruth M. White Scholarship
UMass Amherst Dean's Award

HackUMass IX Hackathon Awards

Won a finalist award and "Best use of Google Cloud" for a Spotify song queue voting app.

PUBLICATIONS IN-PREPARATION

An AI-Driven Acoustic Symphony: Optimized Hydrogel Dewatering Leveraging a Diffusion-model for Music Spectrogram Inpainting

Shuvo, I., Monzón, S., Smith, G., Flores, R., Diaz-Marin, C., Gu, R., Nagle, S., Paradiso, J., Boriskin, V., & Boriskina, S.

Designed and tested an AI framework for improved hydrogel air-water extraction using music generated via Mel-spectrogram Stable Diffusion (Riffusion). Analyzed hydrogel response data to inversely design optimal waveforms.

A Wearable Waterproof System for Long-Range Health Monitoring and Intelligent Distress Signaling

Shuvo, I., Monzón, S., & Boriskina, S.

Designed, 3D printed, and coded an embedded swim-cap device with long-range radio (ESP32 w/ LoRa), IMU, and pulse sensor for real-time machine learning interpretation of swimmer movement and biomedical data. Programmed to transmit, display, and record movement, heart rate, and heart rate variability (HRV) data in real-time with a LoRa receiver.

Musically Interpreting a Simple Generative Transformer

Cosme-Clifford, N., Symons, J., Kapoor, K., Monzón, S., & Wm. White, C.

Explored the aptitudes and failures of a GPT-style model trained on choral-style hymn MIDI files. Encoded chords into Unicode for GPT interpretation and analyzed its learned latent space for implications of AI on music. Investigated existing self-attention methods for music transformers.

CONFERENCES & PRESENTATIONS

Symphony of Dewatering Water Harvesting Hydrogels using AI-Generated Music

[2025 Material Research Society Fall Meeting, Boston](#)

12/2025

I will be presenting my updated Stable Diffusion framework for improved hydrogel water extraction at the MRS Fall conference on December 4th.

Music Generation with Stable Diffusion Inpainting

[Dept. of Multifunctional Metamaterials \(META\) at MIT](#)

04/2025

Presented research on an AI framework I developed to improve hydrogel dewatering using Stable Diffusion inpainting methods.

ADDITIONAL RESEARCH EXPERIENCE

MAGE: Motion-to-Audio Generative autoEncoder

06/2025 - Present

A multi-modal variational auto-encoder (MMVAE) for hand-gesture to audio synthesis.

- LSTM (Long-Short Term Memory) and 1D Causal TCN (Temporal Convolutional Network) blocks for deep learning of sequential hand position data to BRAVE (Bravely Realtime Audio Variational autoEncoder) latent space mapping.
- Uses high FPS video to encode hand gestures for accurate positioning and inference of corresponding conga drum sounds.
- Created a dataset of hand-landmarks and audio encodings from a BRAVE model trained on professional conga drum recordings.
- Used libraries including OpenCV, MediaPipe, and PyTorch.

Music-Emotion Study

Amherst, MA

[Department of Music, UMass Amherst](#)

08/2025 - Present

Exploring human perceptual responses to music features in collaboration with my undergraduate honors thesis advisor.

- Composed two pop music pieces with fixed chord progressions across and variation across pitch, mode, and rhythmic speed.
- Collaborated with musicologists and data scientists on defining experimental test setup and survey metrics, to minimize confounding variables.
- Using EKG, EEG, and GSR to measure the physiological responses of study participants.

Towards Generative Adversarial Audio Synthesis via Latent Interpolation Between Desired Timbral Qualities

Amherst, MA

[Commonwealth Honors College, UMass Amherst](#)

04/2023

My senior year Honors Thesis project exploring Generative Adversarial Networks (GANs) and multi-class classification networks.

- Designed and built a multi-class classification network to categorize sounds by their instruments with 76% accuracy.
- Researched state-of-the-art AI audio generation models including WaveGAN, WaveNet, and NSynth.
- Proposed an AI model to generate musical notes based on sound descriptors using a GAN model.
- Wrote a peer-reviewed 24-page manuscript.

INDUSTRY EXPERIENCE

Software Engineer

Remote

[Cisco Meraki](#)

06/2023 - Present

- Collaborated closely with UX designers, project managers, and fellow engineers to design, develop, and deploy scalable networking solutions for the Meraki Dashboard.
- Technical Lead for the "Power Settings" page of the Meraki Dashboard to distribute power-saving configuration options to Meraki APs.
- Resolved several time-sensitive, million dollar issues for clients across the globe by debugging Meraki's 4 million line codebase.
- Performed several interviews and assisted onboarding software engineers from India and China.
- Designed an LLM for retrieving Slack and Confluence documents to help developers access information during the 1-week Cisco hackathon.

Startup CTO

Boston, MA

[Beamshyft](#)

2025 - Present

[beamshyft.com](#)

A B2B service for selling and shipping low-cost housing supplies directly from manufacturers to real-estate developers.

- Independently designed, developed, and launched a dynamic website with video, catalogue, and scheduling system.
- Improved project clarity by outlining phases in a statement of work for 4 key development phases.
- Adapted development phases to meet business needs.

Software Engineer Intern

Remote

[Cisco Meraki](#)

2023

- Collaborated the UX team to develop the updated UI for the Radio Settings page using Cisco Magnetic design principles.
- Diagnosed high-priority issues to prevent dashboard outages for customers with large networks.
- Solved several mules for improved Meraki Dashboard user experience.

RELEVANT COURSES

Machine Learning (589)

Graduate course

Artificial Intelligence (383)

Software Entrepreneurship (420)

Machine Learning (H589)

Graduate course (Reinforcement Learning)

Uncertainty, Risk, and Decision Making (499CM)

Fundamentals of Music Theory (110)

ACTIVITIES

Swim-team captain, Robotics Club (computer vision), St. Jude Children's Research Hospital fundraising, TKE recruitment chairman

SKILLS

AI & Machine Learning:

PyTorch, Tensorflow, MediaPipe, Multimodal VAE, RAVE, VRNN, Audio/Signal Processing, CLIP, Stable Diffusion

Programming & Systems:

Python, Matplotlib, Librosa, Pandas, Embedded C++, CUDA, MATLAB, React/Next.js, Ruby on Rails, SQL

Design & Fabrication: CAD (Fusion 360), PCB Design (KiCAD), 3D Printing

Creative Practice: Music Production (Ableton Live), Sound Design, Adobe

Research & Interpersonal:

Problem-Solving & Critical Thinking, Creativity & Innovation (Musician, Inventor, Builder), Collaboration & Communication (Research Partnerships, Recruitment Chairman), Leadership & Initiative (Swim Team Captain, Tech Lead), Adaptability & Resilience, Industriousness

LANGUAGES

English (Native)

Spanish (Proficient)